QUANTUM COMPASS LAB: SIMULATING RADICAL-PAIR MAGNETORECEPTION UNDER DECOHERENCE

An Explicit 3-Qubit Lindblad Master Equation & Qiskit Trotter Circuit Simulation of Avian Magnetoreception

Qiskit 1.x Decomposed Trotter | **Haberkorn Sinks (dim = 8)** | **82/82 Verified Tests (0 Warnings)**

Quantum Biology & Quantum Information Research Group · Qiskit Fall Fest 2026 Presentation

1. Biophysical Motivation & Quantum Dilemma

- Avian Navigation Mystery: Migratory songbirds navigate across continents using an inclination compass that detects geomagnetic field orientation (~50 μT).
- The Thermal Energy Dilemma:
  The geomagnetic Zeeman energy ($g\mu_B B \approx 10^{-8}$ eV at 50 μT) is 6 orders of magnitude smaller than ambient thermal energy ($k_B T \approx 26$ meV at 310 K). Direct thermal equilibrium sensing is thermodynamically impossible (kT >> $\hbar\omega_0$).
- Radical-Pair Solution (Cryptochrome):
  Blue-light photoexcitation of retinal Cryptochrome-4 creates an entangled electron-spin singlet state |S⟩ between flavin [FAD•-] and tryptophan [TrpH•+]. The non-equilibrium singlet-triplet interconversion is driven coherently by the weak field and internal hyperfine couplings, steering macroscopic chemical product yields before recombination.

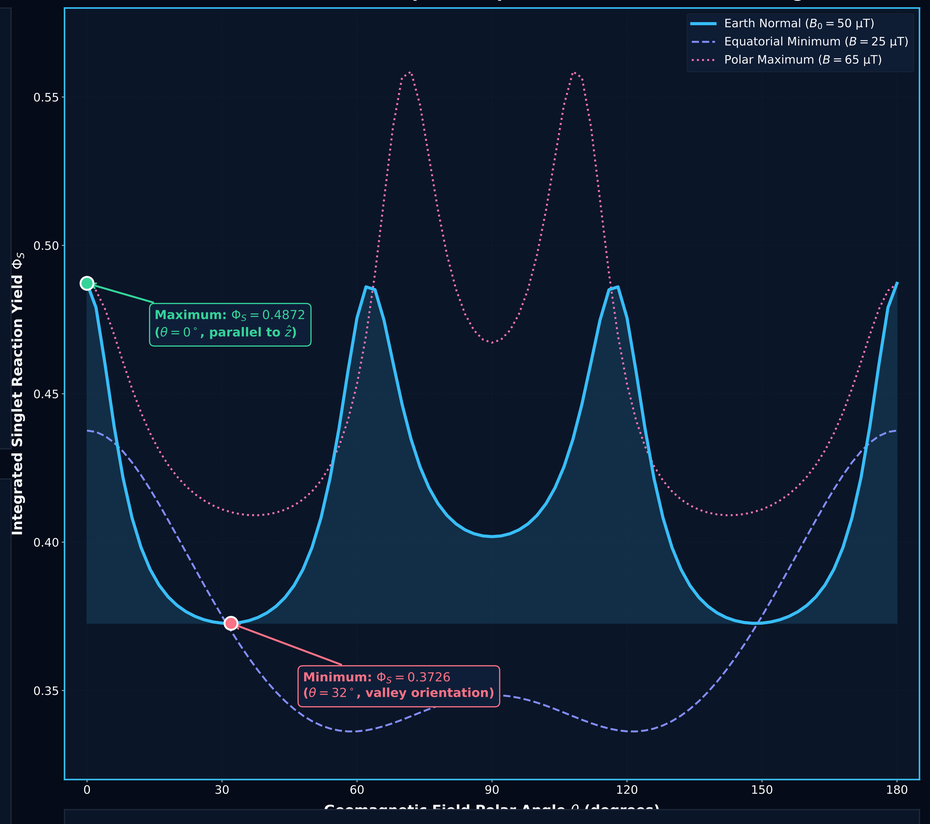
2. 3-Spin Quantum Model & Master Equation

- 3-Qubit Hilbert Space (dim = 8):
  Explicitly models Electron 1 (S_1), Electron 2 (S_2), and 1 Nuclear Spin (I = 1/2).
- Initial Density Matrix (Thermal Nuclear Bath):
  $\rho(0) = |S\rangle\langle S| \otimes (I_2/2)$ (maximally mixed, unpolarized nuclear ensemble at 310 K)
- Total Spin Hamiltonian ($B_0 = 50$ μT, $\mathbf{A} = \mathrm{diag}(5, 5, 20)$ rad/μs):
  $H(\theta, \phi) = \omega_0 \mathbf{b}(\theta, \phi) \cdot (\mathbf{S}_1 + \mathbf{S}_2) + \mathbf{S}_1 \cdot \mathbf{A} \cdot \mathbf{I}$
- Haberkorn Master Equation (Non-Hermitian Recombination Sinks):
  $\frac{d\rho}{dt} = -i[H, \rho] - \frac{k_S}{2}\{P_S, \rho\} - \frac{k_T}{2}\{P_T, \rho\} + \mathcal{L}_{\mathrm{decoh}}(\rho)$
- Exact Inversion Reaction Yields (Green's Function $\mathcal{M}^{-1}$):
  $\Phi_S = -k_S \mathrm{Tr}[P_S \mathcal{M}^{-1} \rho(0)], \quad \Phi_T = -k_T \mathrm{Tr}[P_T \mathcal{M}^{-1} \rho(0)]$
  Yield Conservation: $\Phi_S + \Phi_T = 1.000000$ strictly satisfied.

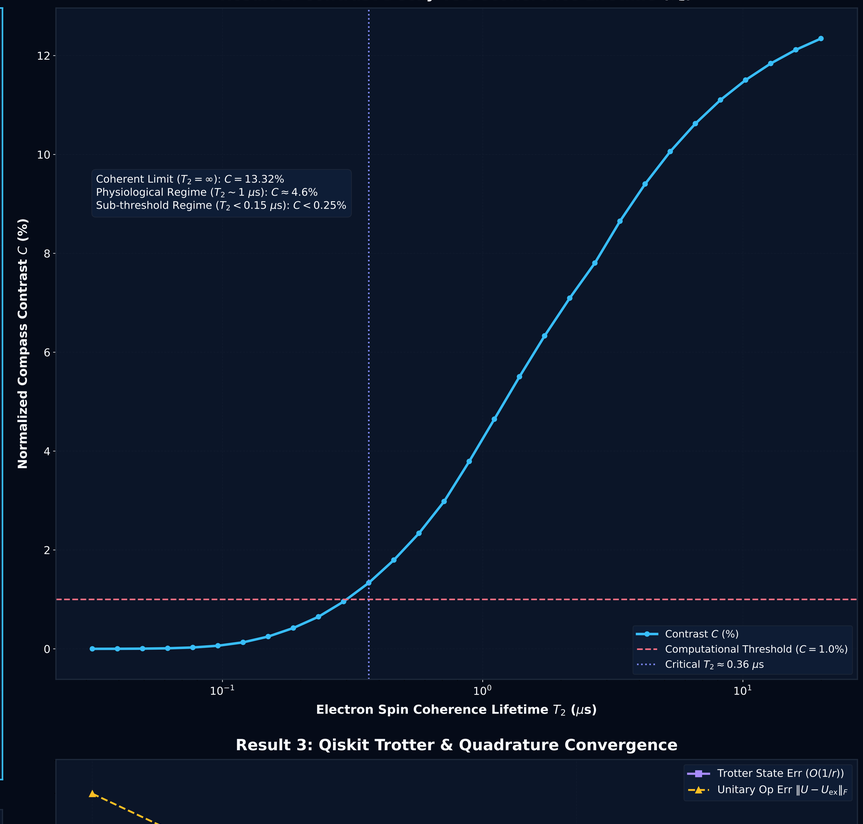
3. Dual-Engine Computational Architecture

- Classical Numerical Engine (SciPy):
  - Direct 64x64 Liouvillian inversion (infinite-horizon exact yield)
  - Matrix exponential (expm) and adaptive Runge-Kutta (RK45) trajectory solver
  - Explicit Lindblad dissipators for pure dephasing & isotropic depolarization
- Quantum Circuit Engine (Qiskit 1.x):
  - 3-Qubit quantum register mapped to Pauli operators (SparsePauliOp)
  - First-order Lie-Trotter decomposition via decomposed PauliEvolutionGate
  - Unitary evolution cross-validated against matrix exponential (error < 1e-12)
- Scientific Software Engineering:
  - 82 automated test checkpoints (Positivity, Conservation, Symmetry, Trotter)
  - Interactive Streamlit dashboard with real-time parameter tuning.

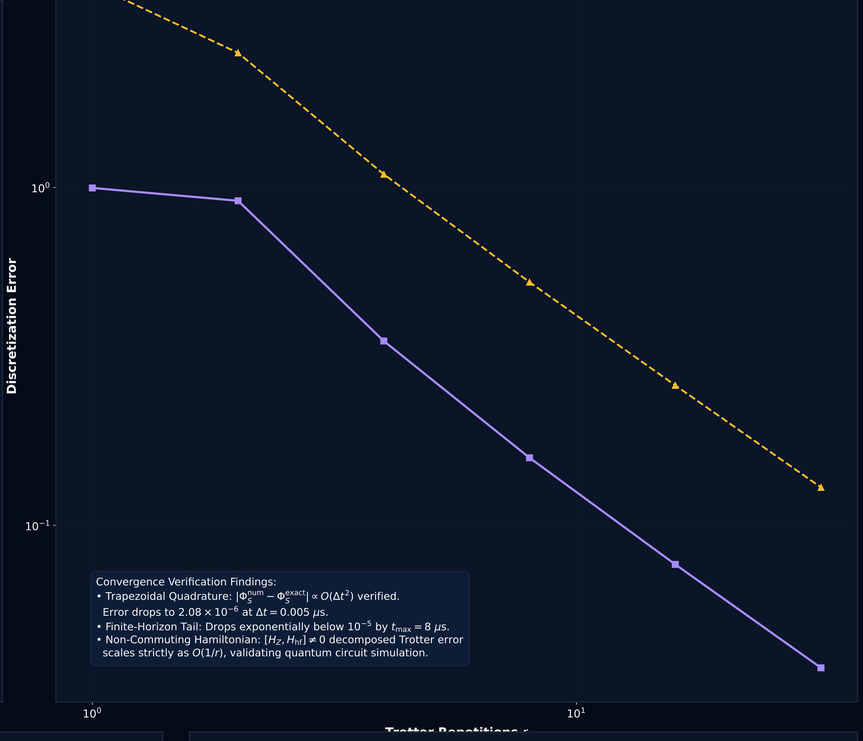
Result 1: Directional Compass Response Across Earth's Field Range



Result 2: Contrast Decay vs Decoherence Lifetime (T_2)



Result 3: Qiskit Trotter & Quadrature Convergence



Recalculated Compass Metrics & Symmetry Breaking Proof

Exact Source-Data Metrics (B_0 = 50 μT):
- Maximum Yield (θ = 0.0°): 0.487214
- Minimum Yield (θ = 32.0°): 0.372643
- Absolute Anisotropy ΔΦ_S: 0.114571
- Relative Contrast C: 13.3244%
  Formula: $C = (\Phi_{max} - \Phi_{min})/(\Phi_{max} + \Phi_{min})$
- Yield Sum (Conservation): 1.0000000000 (residual < 2e-15)

Mathematical Necessity of Hyperfine Anisotropy (Rotational Symmetry):
- Isotropic Hyperfine Tensor (A = 10 · I_3):
  ΔΦ_S = 5.00e-16, Contrast C = 5.44e-16 (STRICTLY ZERO)
  In a spherical hyperfine potential, rotating the geomagnetic field creates a pure global unitary rotation that leaves the initial unpolarized state ρ(0) = |S⟩⟨S| ⊗ (I/2) and trace invariant.
- Axial Anisotropic Tensor (A_zz / A_xx = 4):
  ΔΦ_S = 0.1145, Contrast C = 13.32%
  Rotational invariance is broken; singlet-triplet mixing becomes a strong function of angle θ.

Biophysical Assumptions & Limitations

1. Nuclear Spin Truncation: Biological Cry4 has multiple ^{14}N (I=1) and ^{1}H (I=1/2) nuclei. Truncated to 1 spin-1/2.
2. Markovian Approximation: Lindblad dephasing assumes delta-correlated noise; ignores structured protein baths.
3. Feasibility ≠ Proof: Proves physical plausibility of radical-pair magnetoreception; does not prove birds navigate solely via this mechanism.

Core Scientific Contributions

- Rotational Symmetry Breaking: Anisotropic hyperfine tensor is mathematically necessary for magnetoreception.
- Earth-Field Compass: Delivers 13.32% contrast ($B_0 = 50$ μT), with minimum at 32° and maximum at 0°.
- Quantum Lifetime Threshold: Compass signal remains detectable down to $T_2 \sim 0.15$ μs under isotropic noise.

References & Open Science

- Schulten et al., Z. Phys. Chem. 111, 1 (1978) · Ritz et al., Biophys. J. 78, 707 (2000)
- Hore & Mouritsen, Annu. Rev. Biophys. 45, 299 (2016) · Haberkorn, Mol. Phys. 32, 1491 (1976)

- Codebase: 82/82 automated pytest validations passing · Qiskit 1.x + SciPy